**Schrodinger's Wave Equation For a Hydrogen Atom - By Leo**

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*This was written while I was in Year 12 as a project to help me learn all 4 of my subjects further and with greater rigour - using AI to teach me (not to fully write) the physics and how to solve these equations so that I understand these equations further than just regurgitating definitions. Im writing this from the perspective of teaching a former self these concepts so that I can consolidate the why and how. Although I wouldn't be able to solve this on my own I wanted to understand the why and not just use the maths without knowing what it did*

# 1 Parametric 3D space

Any position in 3D space can be written parametrically in the form of a spherical co-ordinate:

$$x = r\sin\theta\cos\phi, \quad y = r\sin\theta\sin\phi, \quad z = r\cos\theta$$

$$\text{with } r \geq 0, \quad \theta \in [0, \pi], \quad \phi \in [0, 2\pi), \quad \text{and volume element } dV = r^2\sin\theta dr d\theta d\phi$$

Here r is the distance from the origin to a point in 3D space, $\theta$ is the polar angle, measured from $+z$, and $\phi$ is the azimuthal angle in the $xy$ plane, measured from $+x$.

The volume element is the factor you need to multiply to get the new volume when changing the volume, for cartesian co-ordinates this is trivial, $dV = dxdydz$, as you are increasing the size of a cuboid, but with spherical co-ordinates this becomes a bit tricky as you aren't increasing the size of a box.

As a sanity check you could integrate across it:

$$\begin{aligned} V &= \int_0^R \int_0^\pi \int_0^{2\pi} r^2 \sin\theta \, d\phi \, d\theta \, dr \\ &= 2\pi \int_0^R \int_0^\pi r^2 \sin\theta \, d\theta \, dr \\ &= 4\pi \int_0^R r^2 \, dr \\ &= \frac{4\pi}{3} R^3 \end{aligned}$$

which is exactly what you would expect from a sphere.

# 2 The Laplacian

The Laplacian is a differential operator, the Laplacian takes function $f(x, y, z)$ and gives a measure of how much the function curves at each point.
For cartesian co-ordinates the Laplacian is the sum of the second partial derivatives with respect to each co-ordinates, written like this:

$$\nabla^2 f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2}$$

So for example if you had $f(x, y, z) = x^2 + y^2 + z^2$

$$\frac{\partial^2 f}{\partial x^2} = 2, \quad \frac{\partial^2 f}{\partial y^2} = 2, \quad \frac{\partial^2 f}{\partial z^2} = 2 \implies \nabla^2 f = 6$$

Now for spherical co-ordinates the Laplacian is:

$$\nabla^2 f = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}\right) + \frac{1}{r^2\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial}{\partial\theta}\right) + \frac{1}{r^2\sin^2\theta}\frac{\partial^2}{\partial\phi^2}.$$

Which you can derive from doing the Laplacian of $f(x(r,\theta,\phi),\, y(r,\theta,\phi),\, y(r,\theta,\phi))$ using the chain rule.

# 3 The Schrödinger Equation

- A hydrogen atom is a proton with one electron bound to it by electrostatic attraction.
- Classical physics says that the electron should spiral into the proton (accelerating charge radiates energy). It does not, so the classical physics is wrong at this scale.
- Quantum mechanics replaces "the electron is at position $r$" with "the electron's state is described by a wavefunction $\psi(r)$."
- The wavefunction itself isn't directly observable. What is observable is $|\psi(r)|^2$, which gives the probability density of finding the electron at r.
- An orbital is a particular allowed wavefunction, labelled by quantum numbers.

These are adopted postulates from quantum mechanics for our equation, no further assumptions are required.

The Schrödinger Equation is stated as:

$$-\frac{\hbar^2}{2m_e}\nabla^2\psi(r) + V(r)\psi(r) = E\psi(r)$$

Unpacking this term by term:

- $-\frac{\hbar^2}{2m_e}$ is the kinetic energy term, the Laplacian measures the curvature of the wave function - a wave function with more curvature costs more kinetic energy.
- $V(r)\psi$ is the potential energy term. For hydrogen, V is the Coulomb attraction between the electron and proton.
- $E\psi$ is the total energy, where E is the eigenvalue and $\psi$ is the corresponding wavefunction (eigenfunction).

The constants $\hbar, m_e, e, 4\pi\epsilon_0$ clutter our equations, and therefore we adopt atomic units, where
$\hbar = m_e = e = 4\pi\epsilon_0 = 1$ in these units, distances are measured in Bohr radii and energies in hartrees. After these changes our equation simplifies dramatically.
The new equation (given in atomic units) is given as:

$$-\frac{1}{2}\nabla^2\psi + V\psi = E\psi$$

# 4 The Coulomb Potential

The coulomb potential for hydrogen (in atomic units) is:

$$V(r) = -\frac{1}{|r|} = -\frac{1}{r}$$

Here the minus sign means the attraction to the nucleus. The potential depends only on the distance r from the nucleus, not the direction, we can exploit this property later to seperate the equation.

# 5 The Equation To Be Solved

We now finally have the equation:

$$-\frac{1}{2}\nabla^2\psi - \frac{1}{r}\psi(r) = E\psi$$

Which has the following boundary conditions:

- Normalisability: $\int |\psi|^2 dV = 1$, so $|\psi|^2$ is a probability distribution
- Vanishing at infinity: $\psi \to 0$ as $r \to \infty$
- Single-valued: $\psi$ has at most one value at each point

# 6 Manipulating The Equations To Solvable Forms

## 6.1 Substituting

Before we noted that the potential $V(r) = -\frac{1}{r}$ depends only on the distance from the nucleus, not on direction. The atom looks the same from every angle. We will now try a wavefunction of the form $\psi(r, \theta, \phi) = R(r)\Theta(\theta)\Phi(\phi)$

substituting $\psi(r, \theta, \phi) = R(r)\Theta(\theta)\Phi(\phi)$ into our previous Schrödinger equation we recieve:

$$-\frac{1}{2}\left[\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial\psi}{\partial r}\right) + \frac{1}{r^2\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial\psi}{\partial\theta}\right) + \frac{1}{r^2\sin^2\theta}\frac{\partial^2\psi}{\partial\phi^2}\right] - \frac{1}{r}\psi = E\psi$$

subsituting $\psi = R\Theta\Phi$

$$\frac{\partial}{\partial r}(r^2\frac{\partial(R\Theta\Phi)}{\partial r}) = \Theta\Phi\frac{d}{dr}(r^2\frac{dR}{dr})$$

$$\frac{\partial}{\partial\theta}(\sin\theta\frac{\partial(R\Theta\Phi)}{\partial r}) = R\Phi\frac{d}{d\theta}(\sin\theta\frac{d\Theta}{d\theta})$$

$$\frac{\partial^2(R\Theta\Phi)}{\partial\phi^2} = R\Theta\frac{d^2\Phi}{d\phi^2}$$

Ordinary d's replace partials as each function now depends only on one variable. Substituting again:

$$-\frac{1}{2}\left[\Theta\Phi\,\frac{1}{r^2}\frac{d}{dr}\left(r^2\frac{dR}{dr}\right) + \frac{R\Phi}{r^2\sin\theta}\frac{d}{d\theta}\left(\sin\theta\frac{d\Theta}{d\theta}\right) + \frac{R\Theta}{r^2\sin^2\theta}\frac{d^2\Phi}{d\phi^2}\right] - \frac{1}{r}R\Theta\Phi = E\,R\Theta\Phi$$

Divide through by $R\Theta\Phi$

$$-\frac{1}{2}\left[\frac{1}{R}\frac{1}{r^2}\frac{d}{dr}\left(r^2\frac{dR}{dr}\right)+\frac{1}{\Theta}\frac{1}{r^2\sin\theta}\frac{d}{d\theta}\left(\sin\theta\frac{d\Theta}{d\theta}\right)+\frac{1}{\Phi}\frac{1}{r^2\sin^2\theta}\frac{d^2\Phi}{d\phi^2}\right]-\frac{1}{r}=E$$

Now multiply through by $-2r^2$ and rearrange

$$\frac{1}{R}\frac{d}{dr}\left(r^2\frac{dR}{dr}\right)+2r^2\left(E+\frac{1}{r}\right)+\frac{1}{\Theta\sin\theta}\frac{d}{d\theta}\left(\sin\theta\frac{d\Theta}{d\theta}\right)+\frac{1}{\Phi\sin^2\theta}\frac{d^2\Phi}{d\phi^2}=0$$

## 6.2 Seperating r From The Angles

The equation must hold for all points in space. Consider varying r while holding $\theta$ and $\phi$ constant. the first group of terms only vary as r varies, while the further terms are unaffected by r. So for their sum to equal zero regardless of $r$, each group must seperately be constant - and those constants must be equal and opposite

$$\underbrace{\frac{1}{R}\frac{d}{dr}\left(r^2\frac{dR}{dr}\right)+2r^2\left(E+\frac{1}{r}\right)}_{\text{depends only on r}}+\underbrace{\frac{1}{\Theta\sin\theta}\frac{d}{d\theta}\left(\sin\theta\frac{d\Theta}{d\theta}\right)+\frac{1}{\Phi\sin^2\theta}\frac{d^2\Phi}{d\phi^2}}_{\text{depends only on }\theta\phi}=0$$

Introducing the seperation constant $\lambda$:

$$\frac{1}{R}\frac{d}{dr}\left(r^2\frac{dR}{dr}\right)+2r^2\left(E+\frac{1}{r}\right)=\lambda$$

$$\frac{1}{\Theta\sin\theta}\frac{d}{d\theta}\left(\sin\theta\frac{d\Theta}{d\theta}\right)+\frac{1}{\Phi\sin^2\theta}\frac{d^2\Phi}{d\phi^2}=-\lambda$$

## 6.3 Second Seperation of $\theta$ from $\phi$

Multiply the second equation by $\sin^2\theta$:

$$\frac{\sin\theta}{\Theta}\frac{d}{d\theta}(\sin\theta\frac{d\Theta}{d\theta})+\lambda\sin^2\theta+\frac{1}{\Phi}\frac{d^2\Phi}{d\phi^2}=0$$

Now again we see that the first terms depend only on $\theta$ while the further terms only depend on $\phi$. So we can introduce a second seperation constant. We write this as $-m^2$, we write this constant as $-m^2$ as the equation for $\Phi$ will have oscillatory solutions in the form $e^{\pm im\phi}$.
Then:

$$\frac{1}{\Phi}\frac{d^2\Phi}{d\phi^2}=-m^2$$

$$\frac{\sin\theta}{\Theta}\frac{d}{d\theta}(\sin\theta\frac{d\Theta}{d\theta})+\lambda\sin^2\theta=m^2$$

Re-arranging both into solvable forms:

$$\frac{d^2\Phi}{d^2\phi}=-m^2$$

$$\frac{1}{\sin\theta}\frac{d}{d\theta}\left(\sin\theta\frac{d\Theta}{d\theta}\right)+\left(\lambda-\frac{m^2}{\sin^2\theta}\right)\Theta=0$$

### 6.4 Final Solvable Forms

$$(R) \quad \frac{1}{R}\frac{d}{dr}\left(r^2\frac{dR}{dr}\right) + 2r^2\left(E + \frac{1}{r}\right) = \lambda$$

$$(\Theta) \quad \frac{1}{\sin\theta}\frac{d}{d\theta}\left(\sin\theta\frac{d\Theta}{d\theta}\right) + \left(\lambda - \frac{m^2}{\sin^2\theta}\right)\Theta = 0$$

$$(\Phi) \quad \frac{d^2\Phi}{d^2\phi} = -m^2$$

## 7 Solving These New Equations

### 7.1 First ODE

Out of these equations the simplest to solve will be the azimuthal equation for $\Phi(\phi)$
Assume $\Phi = e^{k\phi}$ and differentiate twice:

$$\frac{d\Phi}{d\phi} = ke^{k\phi}, \quad \frac{d^2\Phi}{d\phi^2} = k^2e^{k\phi}$$

$$k^2e^{k\phi} = -m^2e^{k\phi}$$

$$k^2 = -m^2 \implies k = \pm im$$

$$\Phi = Ae^{im\phi} + Be^{-im\phi}$$

Now we must impose boundary conditions. The wave function must be single valued meaning that a point in space cannot have two different probabilities. Since $\phi$ and $2\pi + \phi$ describe the same physical point we require:

$$\Phi(\phi + 2\pi) = \Phi(\phi)$$

Using $\Phi(\phi) = e^{im\phi}$:

$$e^{im(\phi+2\pi)} = e^{im\phi} \implies e^{2\pi im} = 1$$

$$m = 0, \pm1, \pm2, ...$$

$$\boxed{m \in \mathbb{Z}}$$

### 7.2 Normalising This Result

Now we need to fix the constant A by requiring

$$\int_0^{2\pi} |\Phi|^2 d\phi = 1$$

For the complex form $\Phi = Ae^{im\phi}$, this is:

$$\int_0^{2\pi} |A|^2 d\phi = 2\pi|A|^2 = 1 \implies |A| = \frac{1}{\sqrt{2\pi}}$$

Giving:

$$\boxed{\Phi_m(\phi) = \frac{1}{\sqrt{2\pi}}e^{im\phi}, \quad m \in \mathbb{Z}}$$

So we have recieved our first quantum number, m, which controls how the wavefunctions varies as you go around the z axis.

## 7.3 The Second ODE

The first azimuthal equation gave us our first quantum number m. Now we need to solve the polar equation $\Theta(\theta)$ which turns out being much harder.
We need to solve:

$$\frac{1}{\sin\theta}\frac{d}{d\theta}\left(\sin\theta\frac{d\Theta}{d\theta}\right)+\left(\lambda-\frac{m^2}{\sin^2\theta}\right)\Theta=0$$

With $\lambda$ still an undertemined seperation constant meaning, unlike the previous ODE, this has non constant coefficients, and its solutions are not elementary functions. The equation involves both $\sin\theta$ and $\cos\theta$, and derivatives with respect to $\theta$. If we substitute $u=\cos\theta$, trigonometric functions become polynomials which are easier to work with.

$$\frac{d}{d\theta}=\frac{du}{d\theta}\frac{d}{du}=-\sin\theta\frac{d}{du}$$

So

$$\sin\theta\frac{d}{d\theta}=\sin\theta(-\sin\theta\frac{d}{du})=-\sin^2\theta\frac{d}{d\theta}=-(1-u^2)\frac{d}{du}$$

And applying twice like the equation does

$$\sin\theta\frac{d}{d\theta}(\sin\theta\frac{d\Theta}{d\theta})=(1-u^2)\frac{d}{du}[(1-u^2)\frac{dP}{du}]$$

Where $P(u)=\theta(\arccos u)$ Substitute into the equation, divide through by $\sin^2\theta$ and you get:

$$\frac{d}{du}\left[(1-u^2)\frac{dP}{du}\right]+\left(\lambda-\frac{m^2}{1-u^2}\right)P=0$$

This is the associated Legrendre equation which is a standard second-order ODE which solutions are known and we will use.